Lab 4: Sensors for Biomedical Research

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1 Student Outcomes

After completing this laboratory, students will be able to:

1. Interface multiple sensors with an Arduino microcontroller, including orientation, distance, heart-rate, and inertial measurement sensors.
2. Acquire, save, and visualize sensor data in `.txt` or `.csv` format for further analysis using software tools such as Microsoft Excel.
3. Analyze sensor measurements to estimate physical quantities such as orientation, distance, and heart rate, while understanding the effects of noise, calibration, and motion on sensor accuracy.

2 Equipment list

1. Arduino UNO Rev 3 and USB A to B cable (commonly used on printers).
2. 1 Tilt Ball Switch
3. 1 Ultrasonic sensor
4. 1 Photoresistor
5. 1 AD8232 ECG sensor
6. 1 MPU6050 Inertial Measurement Unit
7. DC voltage supply
8. Use any AI tool such as ChatGPT, Gemini, etc. to help you with any problem.

3 Reading

This part of the lab needs to be done before you come to the lab. Assigned reading is listed below

1. Read about tilt switch sensors for orientation https://learn.adafruit.com/tilt-sensor
2. Read about the ultrasonic sensor for measuring distance https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/
3. Read about AD8232 ECG sensor to measure heart rate https://learn.sparkfun.com/tutorials/ad8232-heart-rate-monitor-hookup-guide/all
4. Read about the MPU6050 Inertial Measurement Unit to measure orientation. https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/

4 Exercises

4.1 Saving data to a txt or csv file

The data from sensors will have to be saved on the computer for further processing. This video shows how to save data a .txt or .csv file for processing with Microsoft Excel using an application called Cool Term. https://www.youtube.com/watch?v=j1x54ixEFak. Optionally you can directly send to Microsoft Excel and visualize it there. https://www.youtube.com/watch?v=M-Sr35YIW8o.

4.2 Tilt ball switch, an orientation sensor

Connect a tilt ball switch to give a binary output based on its orientation. Hook the binary switch between any digital pin. Use appropriate code to read their output. Change their orientation and see the output. Answer the question below. What range of inclinations does the tilt sensor switch give an output of 1. Show the circuit and demonstrate the results to the TA.

4.3 Ultrasonic sensor, a distance sensor

This webpage gives the theory behind the ultrasonic sensor and how to hook it up: https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/. Hook up the sensor to the Arduino and calibrate it to read the distance. Demonstrate that the sensor can measure distance and show it to the Teaching Assistant.

4.4 AD8232 ECG sensor, a heart rate measurement sensor

This webpage gives the theory behind the ECG sensor and how to hook it up: https://learn.sparkfun.com/tutorials/ad8232-heart-rate-monitor-hookup-guide/all. Hook up the sensor, collect data, and analyze to answer these questions. From the data, can you predict that humans have a resting heart rate of $\sim 60 - 100$ beats per minute and that heart rate increases and stabilizes with repeated activity (e.g., hand or leg movement)? Demonstrate the end result to the teaching assistant.

4.5 MPU6050 Inertial Measurement Unit to measure orientation.

This webpage gives theory and hookup instructions for MPU6050 IMU: https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/. The Arduino code shows how to get the acceleration and angular velocity (gyro values). Now search and code to see how the IMU can be used to measure orientation. Verify the orientation measurements independently.